EcoExplain-VLM: A Benchmark for Classifying and Explaining Forest Vegetation Change with Vision-Language Models

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Abstract

Vision-language models (VLMs) promise to turn multi-sensor satellite evidence into decisions that people can read and check. We introduce EcoExplain-VLM, a benchmark of 366 forest patches in Kanha Tiger Reserve, India, that pairs matched dry-season Sentinel-2 images (January-February 2022 and 2023) with an evidence card summarising Sentinel-2 indices, Sentinel-1 backscatter, CHIRPS rainfall, ERA5-Land anomalies and NASADEM terrain. Each patch carries a three-class vegetation-change label (decline, stable, increase) derived from NDVI change, and every model is tested on the same 55 held-out patches. A central contribution is a leakage-controlled protocol. With the original card, whose final summary line restates the label, Qwen2-VL-2B reaches 56.4% zero-shot; when that line is removed and all models receive the same numeric table, Qwen2-VL-2B scores 32.7% and the best model, LLaVA-Interleave-0.5B, 43.6%, close to the 34.5% majority baseline. A balanced, label-free LoRA on 16 patches gives 36.4%, whereas a one-number NDVI threshold reaches 94.5%, showing that the evidence is sufficient and that small VLMs do not yet exploit it. For explanations, a structured cite-only prompt makes 75.8% (Qwen2-VL-2B) and 71.1% (LLaVA) of checkable claims agree with the measured changes, and an explanation-tuned Qwen2-VL-2B separates decline from increase (61.8%, significantly above the majority baseline). The benchmark, protocol and code provide a reproducible basis for developing evidence-grounded VLMs for forest monitoring.

Keywords: vision-language models; forest vegetation change; multi-source remote sensing; Sentinel-2; Sentinel-1 SAR; evaluation leakage; LoRA; evidence-grounded explanation; Kanha Tiger Reserve

1 Introduction

India's protected forests are tracked using various satellite products and evaluation methods. Global forest-change maps [3], time-series analysis and radar-based disturbance alerts now effectively detect vegetation changes. These solutions sometimes lack a concise and verifiable explanation of the change, which is important for forest managers and policymakers.For small variations in vegetation, such as those caused by drought or grazing, the evidence is often subtle and distributed across multiple sources.

Vision-language models offer a natural way to provide them. They read images and text together and answer in free language [8, 10], and remote-sensing variants already answer questions about overhead imagery, caption changes between image pairs and ground their answers spatially [7]. A VLM can potentially classify forest changes and explain its predictions using before-and-after images and summarized optical, radar, climate, and terrain data.

Four characteristics of existing models necessitate thorough assessment. Firstly, VLMs have the capability to articulate information that isn't included in the input. Secondly, a coherent explanation doesn't always equate to an accurate one: the reasons provided may not represent how the model derived its conclusion [5]. Thirdly, smaller models are highly influenced by the phrasing of prompts and have consistent response tendencies. A further concern relates to the benchmark itself: if any component of the input discloses the answer, measuring accuracy assesses the ability to retrieve that information rather than the actual intended skill [6]. Consequently, a valuable benchmark for evidence-based VLMs must manage what the model perceives and evaluate its outputs.

This paper presents EcoExplain-VLM, a benchmark and evaluation protocol built for that purpose

in Kanha Tiger Reserve, central India. We ask three questions:

1. Can open VLMs with 0.5-3 billion parameters classify relative vegetation change from before/after images and multi-sensor evidence without task-specific training, and how does the answer depend on what the evidence card contains?

2. Does a small, class-balanced LoRA fine-tune [4] change that ability when the label cannot be read from the input?

3. When prompted to explain their responses, do the models provide justifications that align with the observed changes, and what impact do the arrangement of prompts and fine-tuning on justifications have on that alignment?

Our contributions are fourfold:

1. A reproducible benchmark comprising 366 patches from five data sources, with matched dry-season windows, fixed data splits, and an additional spatial block split.

2. A leakage-controlled evaluation protocol that reruns key experiments without the card's summary line and uses identical inputs across models to quantify its influence on apparent zero-shot performance.

3. An automated claim-verification method that evaluates directional statements in explanations against measured changes, combined with a structured cite-only prompt that achieves 71–76% claim correctness

4. Fine-tuning experiments that establish baselines for future research, including an explanation-tuned model capable of distinguishing forest decline from increase without access to the label.

2 Data and Methods

2.1 Study Area

Kanha Tiger Reserve lies in the Maikal hills of central India, in the Mandla and Balaghat districts of Madhya Pradesh. Its core zone covers 917.44 km^2 and its buffer zone 1,134.36 km^2, 2,051.79 km^2 in total. The forests are dominated by sal (*Shorea robusta*) with moist and dry mixed deciduous forest and grassland meadows. Most rain falls in the June-September monsoon, and January-February is cool, dry and largely cloud-free, which makes it well suited to optical comparison between years.

All processing used a rectangle enclosing the reserve, 80.433°-81.050° E and 22.117°-22.450° N (2,350.7 km^2). Because the rectangle is larger than the gazetted reserve, some patches lie in the buffer zone or just outside the boundary. The 1,200 candidate patch centres from which the dataset was drawn all fall inside the rectangle; the pool contains an empty strip near 80.75-80.85° E whose origin was not recorded.

2.2 Data Sources

2.2.1 Optical and Synthetic Aperture Radar Data

Sentinel-2 Level-2A surface reflectance [1] was taken from the harmonized Earth Engine collection, which removes the offset introduced by the processing-baseline change of 25 January 2022. Scenes with more than 20% cloud were discarded, clouds, cirrus and shadows were masked with the Scene Classification Layer, and per-pixel median composites were formed for T0 (1 January-28 February 2022; 20 scenes) and T1 (the same window in 2023; 23 scenes). From each composite we computed four indices, NDVI [9], EVI, NDMI and NBR:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}}, \tag{1}$$

$$\mathrm{EVI} = 2.5\,\frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + 6\rho_{\mathrm{Red}} - 7.5\rho_{\mathrm{Blue}} + 1}, \tag{2}$$

$$\mathrm{NDMI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{SWIR1}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR1}}}, \tag{3}$$

$$\mathrm{NBR} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{SWIR2}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR2}}}, \tag{4}$$

Change is defined as

$$\Delta I = I_{T1} - I_{T0}, \tag{5}$$

so a negative ΔNDVI means less greenness in 2023. A true-colour RGB chip was rendered for each window as the visual input to the VLMs.

Sentinel-1 GRD C-band backscatter was composited for the same windows from the dominant (descending) orbit direction (15 scenes at T0, 20 at T1; only Sentinel-1A was operating in both windows). The VV and VH channels respond to canopy structure and moisture and are unaffected by cloud. All compositing and export ran in Google Earth Engine [2].

Source	Variables	Resolution
Sentinel-2 L2A	10 bands; NDVI, EVI, NDMI, NBR	10-20 m
Sentinel-1 GRD	VV, VH (dB), descending	10 m
CHIRPS	window rainfall total	≈5.5 km
ERA5-Land	temperature, soil-moisture anomalies	≈9 km
NASADEM	elevation, slope, aspect	30 m

Table 1: The five data sources, all accessed through Google Earth Engine.

2.2.2 Ancillary Environmental Data

Daily rainfall data from CHIRPS was aggregated over each 58-day period. Monthly 2 m air temperature and top-layer soil moisture data were obtained from ERA5-Land, which we then transformed into month-matched anomalies based on a 2015-2021 climatology:

$$A = \tfrac{1}{2} \sum_{m \in \{\text{Jan,Feb}\}} \left(\overline{X}_{m,\text{window}} - \overline{X}_{m,\text{2015-2021}} \right). \tag{6}$$

Elevation, slope, and aspect at a 30 m resolution were provided by NASADEM. While these layers are not included in the label, they are referenced in explanations concerning drought or terrain when appropriate evidence is available. Table 1 provides a summary of the five data sources.

2.3 Preprocessing and Patch Generation

Each patch is the bounding square of a 320 m circle around a candidate center (64 pixels of 10 m), trimmed to the study rectangle. The export uses geographic coordinates (EPSG:4326), with typical patches measuring approximately 70×65 pixels. Each patch exported 15 rasters, totaling 5,490 files. Candidates with masked fractions above 0.15 in either year were removed, whereas selected patches had masked fractions ranging from 0.8% to 2.3%.

To create VLMs, each RGB chip was padded to a square and extended to a 512×512 canvas. Physical layers were stored as arrays on the same canvas. Otsu thresholding of ΔNDVI and ΔNBR identified potential disturbance boxes, which were only preserved for three patches due to a quality policy. Selected patch centers are at least 639 m apart (median nearest-neighbor distance: 1,430 m).

2.4 Change Evidence and Labelling

Let Δ denote the patch-mean NDVI change and $q_{15} = -0.0377$, $q_{85} = +0.0031$ its 15th and 85th percentiles over the 1,200-patch pool. Following the roadmap's recommendation to use region-specific quantiles, the automatic label is

$$\text{class} = \begin{cases} \text{uncertain} & \text{masked} > 0.15, \\ \text{decline} & \Delta \le q_{15}, \\ \text{increase} & \Delta \ge q_{85}, \\ \text{stable} & \text{otherwise.} \end{cases} \tag{7}$$

Uncertain patches were not sampled. The final dataset holds 366 patches: 120 decline, 124 stable and 122 increase. The region as a whole lost a little greenness between the two years (mean $\Delta = -0.016$, s.d. 0.027), so the classes are relative ranks for this reserve and pair of years: "decline" marks the 15% of patches that lost most greenness and "increase" the 15% that fared best. Only 34 patches changed NDVI by 0.05 or more. One annotator assigned a reference class to 19 patches, agreeing with the automatic label on 8 (42%; Wilson 95% interval 23-64%).

Each patch also has a structured evidence card, generated from its rasters, that lists index values and changes, radar means, rainfall, climate anomalies and terrain. In the original pipeline the card ends with a one-line summary, "Inference for …: vegetation_trend=…, candidate_class=…", which restates the card's trend word and the label. Section 2.6.2 describes how the evaluation controls for this line.

2.5 Dataset Construction and Data Splits

Records are stored as JSON Lines without coordinates. The patches were split by class-stratified random sampling (seed 42) into 256 training, 55 development and 55 test patches; the test set holds 18 decline, 18 stable and 19 increase patches, so always predicting "increase" gives a majority baseline of 34.5%. All results below use this test set. Because random splits of spatial data can overstate accuracy, we also provide a spatial block split in which whole blocks of a 5×4 grid are held out (266/46/54 patches).

2.6 Vision-Language Model Evaluation

2.6.1 Models

We compare four instruction-tuned open checkpoints (Table 2): Qwen2-VL-2B-Instruct [10], LLaVA-NeXT-Interleave 0.5B [8], PaliGemma-3B-mix-224 and BLIP-2 OPT-2.7B. Decoding is greedy, and a deterministic parser maps each an-

Model	Image encoder / language model	Size
Qwen2-VL-2B	dynamic-resolution ViT / Qwen2	2B
LLaVA-Interleave	SigLIP / Qwen1.5-0.5B	0.5B
PaliGemma	SigLIP / Gemma	3B
BLIP-2 OPT	ViT + Q-Former / OPT-2.7B	3.8B

Table 2: The four models (Hugging Face checkpoints).

swer to a class, ignoring text that merely repeats the instruction.

2.6.2 Input Protocols

Every model receives the T0 and T1 chips and is asked for one word: `decline`, `increase` or `stable`. Three text inputs were used:

- **Full card** ("Condition C"): the complete evidence card, including the summary line. Qwen2-VL-2B and LLaVA received the full card; PaliGemma, which declines long prompts, received its NDVI line; BLIP-2, which accepts a single image, received the two chips side by side with a short prompt.
- **Evidence-only**: the full card with the summary line and all label words removed.
- **Compact**: a label-free numeric table of the measured changes, identical for every model.

The evidence-only and compact protocols are leak-free [6], and the compact protocol is also input-equal across models. A script verifies that no leak-free prompt contains the label. As reference points we report the majority baseline and a threshold rule on the card's ΔNDVI value, with cut-offs either fixed at ± 0.02 or fitted on the training split.

2.6.3 Parameter-Efficient Fine-Tuning

We fine-tune Qwen2-VL-2B with LoRA [4], which adds a trainable low-rank update to frozen weights,

$$W = W_0 + \frac{\alpha}{r} BA, \tag{8}$$

with rank $r = 8$, $\alpha = 16$ and dropout 0.1 on all attention and MLP projections. The base model is stored in 4 bits and trained with AdamW (learning rate 5×10^{-5}, weight decay 0.05, gradient accumulation over four steps) on an NVIDIA GTX 1080 Ti. Two adapters are reported.

The **class LoRA** is trained for one epoch on 16 training patches drawn evenly across classes (6 decline, 5 stable, 5 increase) with the evidence-only card at training and test time. The **explanation LoRA** follows a teacher-student design: Qwen2-VL-2B wrote a JSON justification for each training patch from the compact table, and a justification was kept only if its class was correct, it contained at least two checkable claims and all of them agreed with the measured changes. Of 256 justifications, 28 passed; the 16 that fell within the first 128 training patches were used for one epoch.

2.6.4 Evaluation Metrics

Classification is scored by accuracy and macro-F1,

$$\mathrm{Acc} = \frac{\#\mathrm{correct}}{N}, \qquad \text{macro-F1} = \frac{1}{3} \sum_{c} F1_c, \tag{9}$$

with Wilson 95% intervals,

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}, \tag{10}$$

and paired comparisons by McNemar's exact test.

Explanations are scored automatically. The scorer finds each indicator (NDVI, EVI, NDMI, NBR, green cover, VV, VH, rainfall, soil moisture, temperature) and its nearest direction word, converts the measured change into a direction with an indicator-specific dead-band τ (0.02 for the indices),

$$\mathrm{dir}(\Delta) = \begin{cases} \text{stable} & |\Delta| < \tau, \\ \text{increase} & \Delta \geq \tau, \\ \text{decrease} & \Delta \leq -\tau, \end{cases} \tag{11}$$

and counts a claim as correct when the two directions match. We report the share of correct claims, the share of explanations with at least one incorrect claim, and self-consistency (whether the dominant claimed direction agrees with the model's own class). This measures consistency with the evidence rather than faithfulness to the model's internal reasoning [5]; ROUGE-L is reported only as a measure of copying.

2.7 Experimental Workflow

Figure 1 summarises the workflow in six steps: (1) study design with matched windows; (2) collection of the five sources; (3) compositing and export in Earth Engine; (4) image processing, labelling and evidence cards; (5) freezing of the dataset and splits; and (6) three evaluation tracks on the same 55 test patches: zero-shot classification (Track A), class LoRA (Track B) and explanations (Track C).

Quantity	Value
Candidate pool / final patches	1,200 / 366
Decline / stable / increase	120 / 124 / 122
Train / development / test	256 / 55 / 55
Test decline / stable / increase	18 / 18 / 19
Majority baseline (test)	34.5%
Spatial split train / dev / test	266 / 46 / 54
Mean ΔNDVI (s.d.)	−0.016 (0.027)
Patches with $\lvert\Delta\text{NDVI}\rvert \geq 0.05$	34
Human agreement	8 / 19 (42%)

Table 3: EcoExplain-VLM at a glance.

The development set was used only for prompt wording, the closed-set probe's bias and checkpoint selection.

3 Results and Discussion

3.1 Dataset Characteristics

Table 3 summarises the benchmark. The optical indices move together: ΔNDVI correlates with ΔNBR, ΔNDMI and ΔEVI at 0.80, 0.77 and 0.78. January-February 2023 was markedly drier than 2022: mean window rainfall fell from 69.1 to 25.1 mm, the temperature anomaly rose from -1.52 to $+0.88$ °C and the soil-moisture anomaly fell from $+0.024$ to -0.043 m^3/m^3; VV and VH backscatter fell by 0.47 and 0.48 dB. These climate and radar signals are nearly identical across the three classes, so they describe the year rather than individual patches, a useful property when judging whether an explanation's causal claims are supported. Figure 2 shows the label construction.

3.2 Zero-Shot Model Performance

Table 4 reports Track A in two blocks. With the full card, Qwen2-VL-2B reaches 56.4% (95% interval 43.3-68.6%), the highest zero-shot score, while the other models stay near the 34.5% majority baseline. The leak-free protocol explains this result. Removing the summary line lowers Qwen2-VL-2B to 32.7% under both the evidence-only card and the compact table, where it answers "stable" for every patch (Figure 3). On the patches where the two Qwen runs disagree, the full card is right 18 times and the compact table 5 times ($p = 0.011$). The trend words that Qwen followed with the full card appear only in the summary line, so its full-card score reflects that line rather than the measurements.

The equal-input comparison is also informative. Given the same compact table, LLaVA-Interleave-0.5B improves from 32.7% to 43.6%, recognising 16 of 18 decline and 8 of 19 increase patches. Its gain over the baseline is not yet significant at $n = 55$ ($p = 0.44$), and it never predicts "stable", but the change in behaviour shows that a cleaner, numeric input helps the smaller model use the evidence.

Model (input)	Acc.	95% CI
Leak-free		
Qwen2-VL-2B (evidence-only)	32.7	21.8-45.9
Qwen2-VL-2B (compact)	32.7	21.8-45.9
LLaVA-0.5B (compact)	**43.6**	31.4-56.7
PaliGemma-3B (compact)	not available[a]	
BLIP-2 (compact)	no parsable answer	
Full card (includes summary line)		
Qwen2-VL-2B	56.4	43.3-68.6
PaliGemma-3B (NDVI line only)	36.4	24.9-49.6
LLaVA-0.5B	32.7	21.8-45.9
BLIP-2 (images only)	32.7[b]	21.8-45.9
Reference points		
Majority class	34.5	23.4-47.7
ΔNDVI rule, ±0.02	63.6	50.4-75.1
ΔNDVI rule, fitted on train	94.5	85.1-98.1

Table 4: Zero-shot accuracy (%) on the 55 test patches. [a]Gated model, not loaded in the leak-free run. [b]Produced by an echoed prompt under the original parser.

The reference points place these results in context. A rule that reads the single ΔNDVI value on the card and applies two cut-offs fitted on the training split classifies 94.5% of the test patches correctly. The information needed for the task is therefore present on every card; what the small VLMs lack is the ability to extract and compare it. This gap is a clear, measurable target for future models.

3.3 Effect of Fine-Tuning

The balanced, label-free class LoRA reaches 36.4% (95% interval 24.9-49.6%; Figure 4c). After one epoch on 16 examples the adapter answers "stable" for 53 of 55 patches, which is close to the zero-shot behaviour on label-free inputs ($p = 0.50$) and to the majority baseline ($p = 1.0$). Earlier exploratory adapters trained with the full card reached 67.3% on 16 patches and 100% on 256 patches, by learning to read the label from the summary line (Figure 4b; Appendix A); the leak-free protocol correctly flags these as shortcut learning. The balanced run therefore establishes a clean starting point: 16 examples are not enough to teach the task, and the natural next step is a larger, balanced, label-free fine-tune with early stopping.

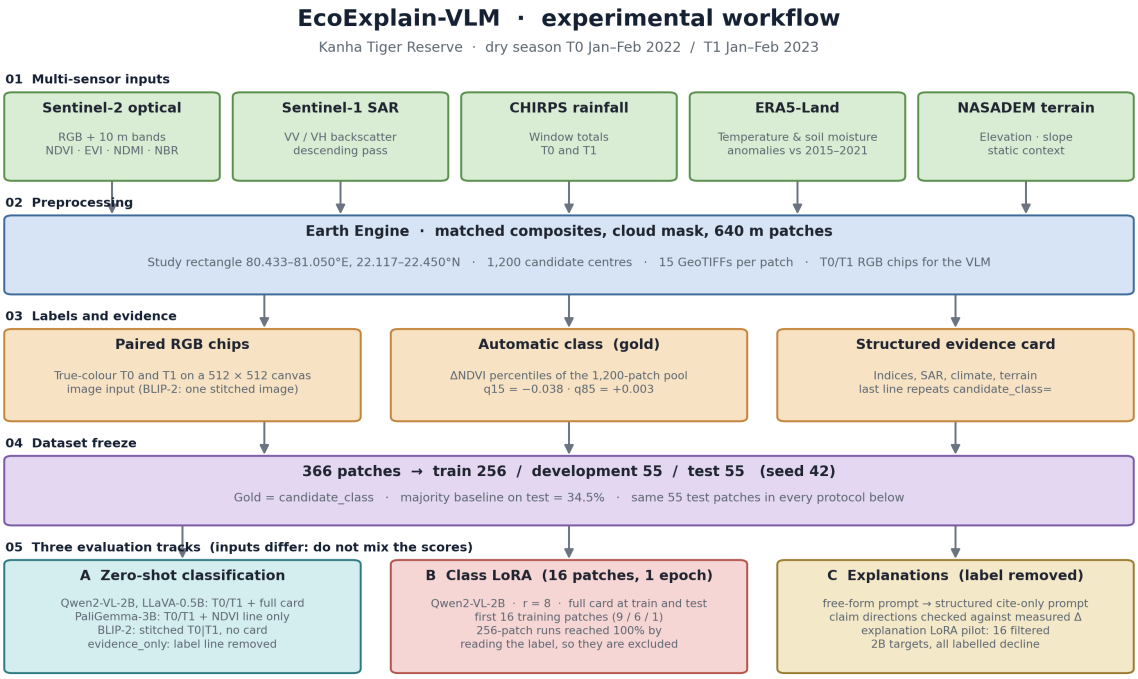


Figure 1: The EcoExplain-VLM workflow. After the dataset is frozen, three evaluation tracks run on the same 55 test patches and are scored separately.

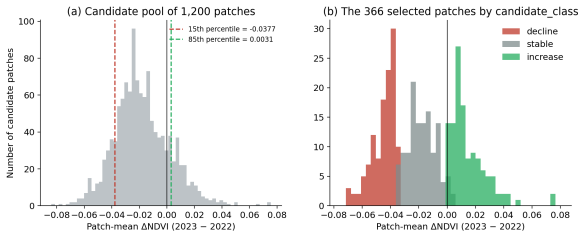


Figure 2: (a) Patch-mean ΔNDVI over the 1,200-patch pool with the 15th and 85th percentiles. (b) ΔNDVI of the 366 selected patches by class.

3.4 Error Analysis

Errors concentrate where the labels are weakest. Because the region became slightly less green overall, many "increase" patches changed NDVI by less than 0.011, and with the full card Qwen2-VL-2B labelled all 19 of them "stable". Each model also shows a characteristic default answer in the leak-free setting ("stable" for Qwen2-VL-2B and the class LoRA; mostly "decline" for LLaVA). A closed-set probe that scores each class word and corrects these preferences with a development-set bias raises LLaVA to 60.0% (Appendix C), which points to calibration as a promising remedy.

Part of the disagreement lies in the labels themselves. The card's own trend words, computed with fixed thresholds, disagree with the percentile label on 108 of 366 patches, and the human check confirmed only 3 of 10 automatic "increase" labels. Some "errors" are thus borderline cases rather than model failures, and the test labels were not adjusted after seeing any model output.

3.5 Explanation Quality

Track C asked the models to justify their answers using the label-free compact table (Table 5, Figure 5). Under a free-form prompt, PaliGemma and BLIP-2 gave no checkable reasons and LLaVA's claims were correct 56.7% of the time.

The structured cite-only prompt lists the facts the model may use, states an NDVI decision rule, and asks for a one-sentence justification. It raises claim correctness to 75.8% for Qwen2-VL-2B and 71.1% for LLaVA, with self-consistency of 100% and 90.7%. Most claims still flagged as incorrect (35 of 38 for Qwen, 35 of 37 for LLaVA) describe true changes smaller than the scorer's dead-band rather than invented facts. The justifications largely restate the supplied facts and Qwen's class follows the stated rule on 53 of 55 patches, which makes the explanations reliable but not yet interpretive.

The explanation LoRA offers the most encouraging classification signal in the study. Working from the label-free table without any stated rule, it classifies 17 of 18 decline and 17 of 19 increase

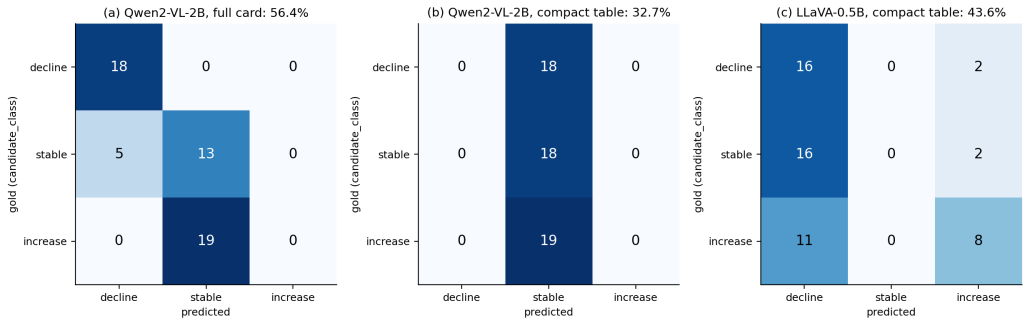


Figure 3: Confusion matrices on the test set. (a) Qwen2-VL-2B with the full card; (b) Qwen2-VL-2B with the label-free compact table; (c) LLaVA-Interleave-0.5B with the compact table.

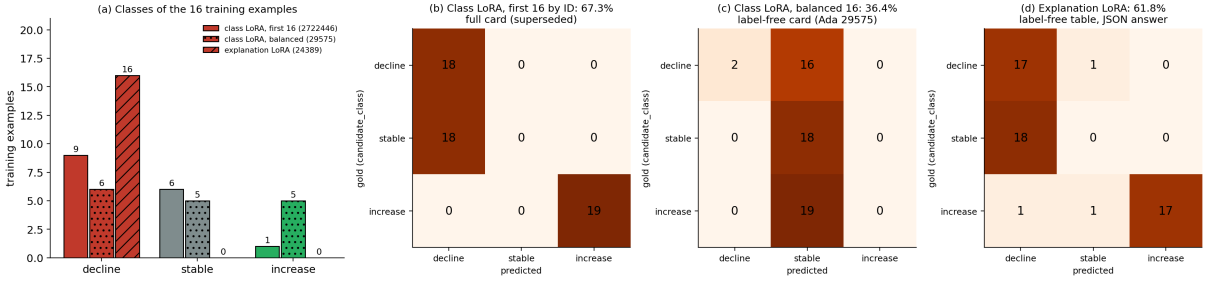


Figure 4: Fine-tuning. (a) Class mix of the 16 training examples of each adapter. (b) Exploratory class LoRA trained and tested with the full card. (c) Balanced, label-free class LoRA. (d) Explanation LoRA, evaluated in explanation mode.

Setting	Class acc.	Correct claims	Self-cons.
Free-form, LLaVA	29.1	56.7	10.3
Structured, Qwen2-VL	63.6	**75.8**	100
Structured, LLaVA	45.1[a]	**71.1**	90.7
Explanation LoRA, Qwen2-VL	**61.8**	50.5	96.4

Table 5: Explanation results (%) on the test set with the label-free table. [a]Of 51 answered patches.

patches correctly (61.8%, interval 48.6-73.5%), significantly above the majority baseline (McNemar: 17 patches gained, 2 lost, $p < 0.001$). It does not yet recognise "stable" patches, and its claim correctness (50.5%) is below that of structured prompting, reflecting a small teacher set (16 examples, all labelled decline). Because the zero-shot counterpart of this setting was run on only three patches, the relative contributions of the prompt and of the fine-tuning remain to be separated; doing so is a priority for follow-up work.

3.6 Comparison with Existing Approaches

Map-oriented change-detection methods provide excellent wall-to-wall disturbance maps [3], and our threshold baseline confirms how much information a single spectral index carries for this task. Remote-sensing language datasets and domain-adapted VLMs [7] have shown that imagery and language can be combined effectively. EcoExplain-VLM builds on these advances by adding a new dimension: it tests whether a general VLM can reason over the same multi-sensor evidence an analyst would use, controls the input for label leakage, and verifies every directional claim against measured values. CLIP-style encoders offer efficient image-pair scoring, and EcoExplain-VLM complements them by evaluating the written justification that makes a decision transparent and checkable.

4 Conclusions

EcoExplain-VLM combines matched dry-season Sentinel-2 and Sentinel-1 data, climate and terrain context and a three-class vegetation-change label into a reproducible benchmark for Kanha Tiger Reserve. Its leakage-controlled protocol proved essential: the strongest zero-shot result with the original card (Qwen2-VL-2B, 56.4%) came from the

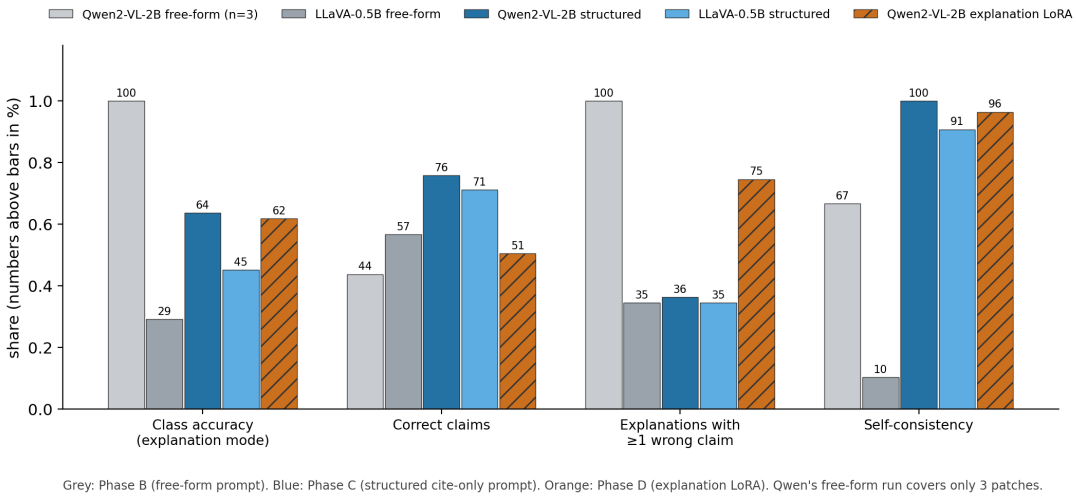


Figure 5: Explanation scores for the free-form prompt (grey), the structured cite-only prompt (blue) and the explanation LoRA (orange). The free-form Qwen2-VL-2B run covers three patches only.

card's summary line, and with that line removed and inputs equalised, small open VLMs remain near the majority baseline, with LLaVA-Interleave-0.5B best at 43.6%. A one-number rule reaching 94.5% shows that the evidence is sufficient, giving future models a clear target. For explanations, a structured cite-only prompt makes more than 70% of checkable claims agree with the measured changes, and an explanation-tuned Qwen2-VL-2B separates decline from increase without access to the label. Together, the dataset, protocol and scoring tools give the community a practical way to measure progress towards VLMs whose decisions about forest change are both correct and verifiable.

Limitations

Image input. The models only saw plain colour (RGB) images of the forest, which look dark and almost the same in both years. They were never shown the NDVI change maps or false-colour images, which show the change much more clearly. Giving the models these more informative images is a simple way they could perform better.

Labels. Our labels come from a simple rule on NDVI change between two dry seasons, not from confirmed field data. Only one person checked 19 patches by hand. Having several experts check more patches, and measuring how often they agree (Cohen's κ), would make the labels more reliable.

Fine-tuning. Each fine-tuned model saw only 16 training examples, once. Training on more examples with balanced classes, and using a stronger model to write the training explanations, should give a clearer picture of what fine-tuning can achieve.

Evaluation tools. Our explanation checker looks for key words such as "NDVI decreased". It cannot understand negative sentences such as "NDVI did not decrease", and it treats very small changes as no change. Having people rate the explanations would add a useful second check. The numbers on each evidence card are also slightly too low (by about 8%), because empty borders were included when averaging; this is easy to fix.

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Appendix

A Fine-Tuning Details

Table 6 lists every class-LoRA run. The two full-card runs on 256 patches reached 100% with a training loss near zero, consistent with reading the label from the summary line; the 16-patch full-card run used the first 16 training patches by identifier (9 decline, 6 stable, 1 increase) and labelled every stable patch "decline". These runs motivated the label-free, balanced protocol used for the reported adapter.

Run	Train	Input (train / test)	Test
Reported, balanced	16	label-free / label-free	36.4
Exploratory	16	full card / full card	67.3
Exploratory	256	full card / full card	100
Exploratory	128	full card / images only	32.7

Table 6: Class-LoRA runs (accuracy in %). All use $r = 8$, $\alpha = 16$, one to three epochs, 4-bit Qwen2-VL-2B.

B Prompts and Output Parsing

Qwen2-VL-2B and LLaVA received both images, protocol-specific card text, and a one-word classification instruction. PaliGemma used a text prompt identifying the T0 and T1 images, while BLIP-2 received side-by-side images and a short classification request. In the leak-free setting, BLIP-2 also received a compact evidence table. The explanation prompt specified permitted facts, the ± 0.02 NDVI threshold, and a one-sentence justification. The parser extracted the first valid class label or synonym after removing echoed instructions.

C Additional Results

A closed-set probe that scores each class word by its log-probability, with a four-panel image, a stated NDVI rule and class biases fitted on the development set, reached 60.0% for LLaVA-Interleave-0.5B (it never predicted "stable"), against 32.7% for PaliGemma and BLIP-2. In an earlier label-free one-word run on 2 October, LLaVA scored 36.4%. Under the full card, a 128-patch adapter tested on images alone scored 32.7%. With $n = 55$, all intervals are about ± 13 percentage points wide, so per-class counts (about 18 patches per class) support only coarse error analysis.